Selection Sort

AlgoExpert · Easy · Sorting

The problem

- Repeatedly select the minimum of the unsorted suffix and swap it to the front.

Intuition

- Repeatedly select the smallest remaining element and swap it into the next position.

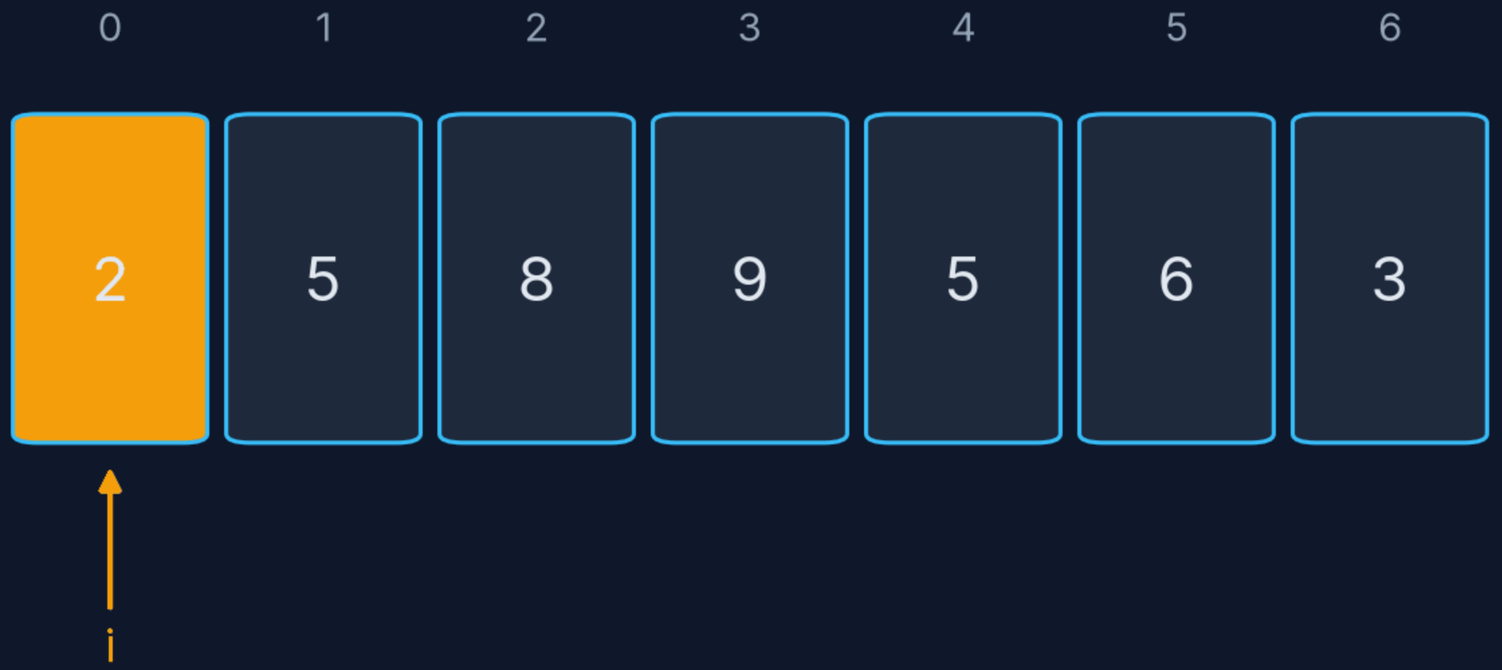
Brute force vs optimal

- Always $O(n^2)$ comparisons, even if sorted
- At most $n-1$ swaps (useful when writes are expensive)
- Use an $O(n \log n)$ sort in practice

Approach

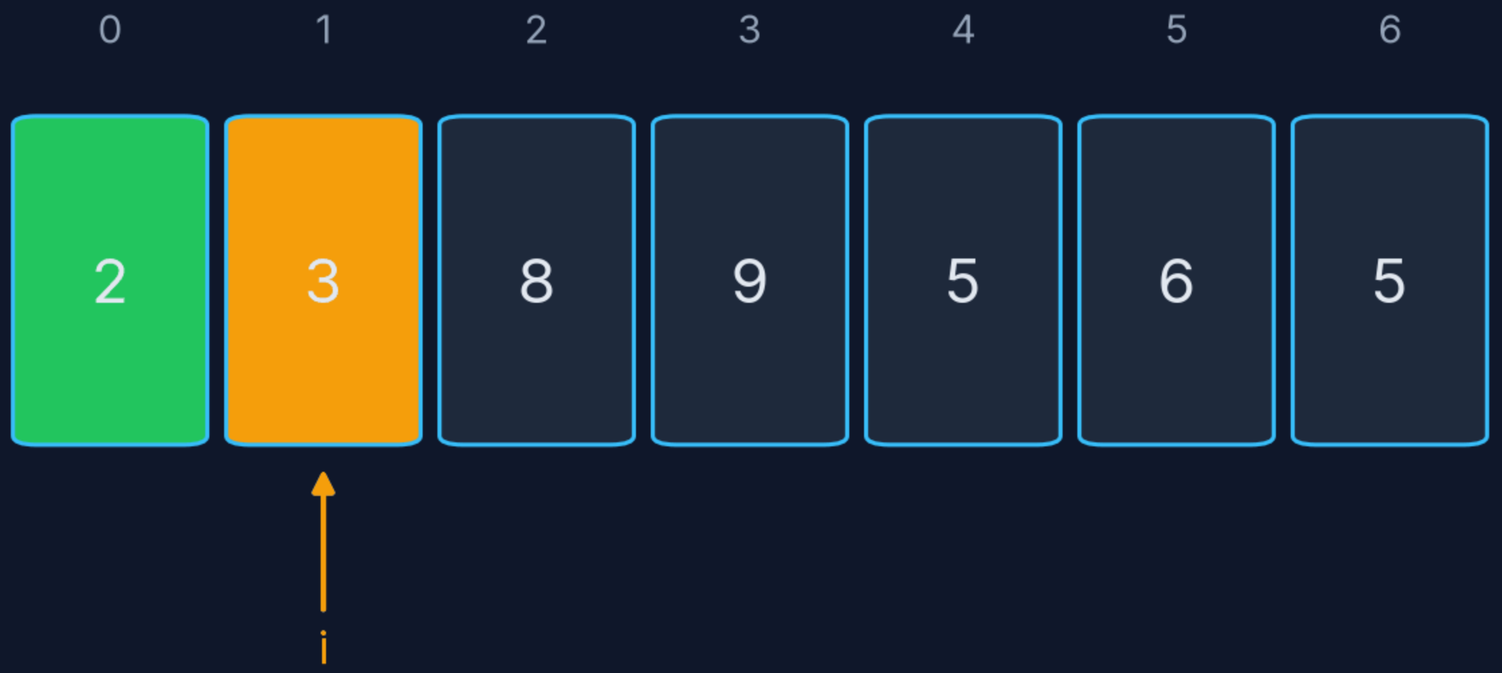
- 1. For each position
- 2. Find the minimum in the unsorted suffix
- 3. Swap it into place

Worked example



Min of suffix = 2 → swap into index 0

Worked example



Min of suffix = 3 → swap into index 1

Worked example



Min of suffix = 5 → swap into index 2

Worked example



Min of suffix = 6 → swap into index 4

Worked example



Min of suffix = 8 → swap into index 5

Java solution

```java
public class SelectionSort {
    public static int[] selectionSort(int[] a) {
        for (int i = 0; i < a.length - 1; i++) {
            int min = i;
            for (int j = i + 1; j < a.length; j++) if (a[j] < a[min]) min = j;
            int t = a[i]; a[i] = a[min]; a[min] = t;
        }
        return a;
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(1)

Edge cases & pitfalls

- Not stable (long-distance swaps)
- Stop at n – 1
- Track the index of the min, not the value

Interview tips & follow-ups

- Compare with bubble and insertion sort
- Follow-up: make it stable (shift instead of swap)

Recap

- Repeatedly select the smallest remaining element and swap it into the next position.
- Complexity: O(n^2)